



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Complex Variables & Laplace Transform (MAT215)

Laplace Transform Using Table

First Translation Theorem, Transform of $t^n f(t)$

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CONDUCTED BY

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What is Laplace Table?

Laplace Transform

The Laplace transform of a function $f(t)$, denoted by $\mathcal{L}\{f(t)\}$, is defined as

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt,$$

where s is a complex number and the integral converges.

Linearity of Laplace Transform

Theorem

Let $f(t)$ and $g(t)$ be functions such that $\mathcal{L}\{f(t)\}$ and $\mathcal{L}\{g(t)\}$ exist. Then, for any constants c , we have

$$\mathcal{L}\{cf(t)\} = c \mathcal{L}\{f(t)\},$$

$$\mathcal{L}\{f(t) + g(t)\} = \mathcal{L}\{f(t)\} + \mathcal{L}\{g(t)\}.$$

Laplace Transform Using Standard Table

② Problem

Evaluate

$$\mathcal{L}\{3e^{2t}\}$$

using the standard table.



② Problem

Evaluate

$$\mathcal{L}\{4t^3 - e^{-t}\}$$

using the standard table.



🔍 Problem

Evaluate

$$\mathcal{L}\{(t^2 + 1)^2\}$$

using the standard table.



 Problem

Evaluate

$$\mathcal{L}\{7 \sin 2t - 3 \cos 2t\}$$

using the standard table.



 Problem

Evaluate

$$\mathcal{L}\{7 \sinh 2t - 3 \cosh 2t\}$$

using the standard table.



Important Trigonometric Identities

Compound Angle Identities

$$\sin(A + B) = \sin A \cos B + \cos A \sin B,$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B,$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B.$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

Product-to-Sum Identities

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B),$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B),$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B).$$

Negative Angle Identities

$$\sin(-A) = -\sin(A),$$

$$\cos(-A) = \cos(A).$$

 Problem

Evaluate

$$\mathcal{L}\{3 \sin 7t \cos 4t\}$$

using the standard table.



? Problem

Evaluate

$$\mathcal{L}\{4 \cos 3t \cos 5t\}$$

using the standard table.



 Problem

Evaluate

$$\mathcal{L}\{4 \cos 3t \cos 7t\}$$

using the standard table.



First Translation Theorem

💡 First Translation Theorem

Let $f(t)$ be a function such that $\mathcal{L}\{f(t)\}$ exists and $F(s) = \mathcal{L}\{f(t)\}$. Then, for any constant a , we have

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a).$$

🔍 Problem

Evaluate

$$\mathcal{L}\{t^3 e^{5t}\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{5e^{-3t} \sin 4t\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{e^{-t}(3 \sinh 2t - 5 \cosh 2t)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{2e^{3t} \sin 5t \sin 3t\}$$



🔍 Problem

Evaluate



$$\mathcal{L}\{3e^{3t} \cos 5t \cos 7t\}$$

🔍 Problem

Evaluate

$$\mathcal{L}\{4e^{3t} \sin 5t \cos 3t\}$$



Laplace Transform of the form $t^n f(t)$

Theorem

Let $f(t)$ be a function such that $\mathcal{L}\{f(t)\}$ exists and $F(s) = \mathcal{L}\{f(t)\}$. Then, for any positive integer n , we have

$$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s),$$

where $F^{(n)}(s)$ denotes the n -th derivative of $F(s)$ with respect to s .

🔍 Problem

Evaluate

$$\mathcal{L}\{t^2 \sin 2t\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{te^{-2t} \sin 3t\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{te^{2t}(2 \sinh 3t + 5 \cosh 4t)\}$$



🔍 Problem

Evaluate



$$\mathcal{L}\{3t \sin 7t \cos 5t\}$$

🔍 Problem

Evaluate

$$\mathcal{L}\{4t \cos 3t \cos 5t\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{7te^{-4t} \sin 5t \sin 3t\}$$



Thank You!

We'd love your questions and feedback.

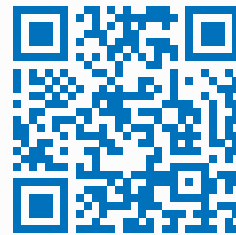
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(Lectures, walkthroughs, and course updates)



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References

- [1] Murray R. Spiegel, *Schaum's Outline of Complex Variables*, 2nd Edition, McGraw-Hill, 2009.
- [2] Dennis G. Zill, *A First Course in Differential Equations with Modeling Applications*, 11th Edition, Cengage, 2018.